

Project 2

Data feature extraction.

Algo.
Results.

multiclass classification of
epileptic seizures using ML.

electroencephalography.

EEG signals from Bonn university dataset
five sets (A-E).

used to record electrical activity in the brain using
electrodes placed on the scalp.

numpy → for numerical operations

os → file operations reading files from folders

sklearn (scikit-learn) → open source ML library in Python.

offers various features — supervised learning
un —

model selection →

preprocessing. —

feature engineering —

metrics & evaluation →

scipy.signal.welch →

power spectral density calc

describes how power is
distributed over diff frequencies

why welch? → handles noise well

→ computationally faster than traditional
periodogram method.

pywt

for wavelet transforms.

shows both time & freq
details.

helps us analyze signals at
different scales & time periods.

why? → unlike Fourier transform (shows freq content).
shows both

— good for non-stationary signals

Dataset divided in 5 set (A $\rightarrow$ E).

500 - 600 signals.

100 in each.

duration - per signal $\rightarrow$ 23.6 sec.

sampling freq $\rightarrow$ 173.61 Hz. 4097 data points per file

	Brain state	cond.	recording loc.
A	Awake	normal EEG (eye open)	extracranial
B	—	— (eye closed)	—
C	seizure free	epileptic brain not during seizure	intracranial
D	—	—	—
E	seizure	during actual seizure	—

extracranial $\rightarrow$ outside skull (scalp electrodes)

intra $\rightarrow$ inside brain

• txt file.

C $\rightarrow$ healthy part of brain
D $\rightarrow$ seizure zone
(epileptogenic zone)

labeling data $\rightarrow$ label 0 $\rightarrow$ set A

1 B
2 C

How spread out the values in a dataset are

Data Normalization

Scaler = StandardScaler $\rightarrow$ mean of 0 & SD=1

Normalization improves ML model performance

- min-max scaling

- z score normalization

- PSD
- FFT
- wavelet

(why) in next page.

Why? - data loss less.
- efficient & fast

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Feature extraction

uses Welch method $\rightarrow$ PSD (power spectral density)

How power is distributed over diff freq.

PSD captures freq-based features from EEG signals

EEG signal processing to analyze brainwave activity.

How Welch method works?

1) split the signal into overlapping segments.

Signal - break into parts

2) Apply a window function to each segment

Hamming window

$\rightarrow$ reduce edge effects.

(best balance b/w
freq resolution &
noise reduction)

Hamming - smoother transition.

Blackman - reducing noise & resolution.

3) compute FFT (fast Fourier transform) for each segment.

convert

time domain to freq domain

convert signals to
freq components.

4) Average the power

convert

power of each freq

is avg.

$\rightarrow$ reduce noise &

improve accuracy.

signals (time domain
(amplitude vs time))

freq domain
(amplitude vs
freq.)

helps analyze diff freq present in signal
- detect brainwave frequencies (α , β , γ , etc).

split Training 80% Testing 20%.

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Classifiers

- Random forest
- SVM
- Naive Bayes
- Decision Tree.

Why PSD over other?

PSD provides better freq resolution for stationary ECG signals.

- more robust to noise & variability in ECG signals.
- computationally efficient compared to wavelet transforms.
- PSD captures imp. ECG features (a, B, Y, S, T) more efficiently.

Random forest → handles non linear ECG data.
reduces overfitting.
good with missing data.

Support vector machine → great for high dimensional ECG data.

efficient for binary classification

↳ classified into seizure/non seizure states

Naive Bayes → works well with independent ECG features.
- fast & efficient.
requires less training time & data.

Assumes that features are independent

decision tree → you can see how each & every feature contributes to the classification.

- handles both linear & non-linear data.
- (does not assume a linear boundary)
- good for small datasets

Hyperparameter tuning

Grid Search CV → find the best hyperparameter for a ML model by systematically searching over a predefined grid of parameter values.

- trying all comb^s of hyperparameters
- evaluating comb^s with cross-validation
- select best model based on highest performance score

Model evaluation

Accuracy, precision, recall, f1-score

$F1\ score \rightarrow \text{balance precision \& recall}$

$F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$

Multi class classification

- diffⁿ disease types need diff. treatment.
- improve diagnosis
- tailored patient care.

Why this?

- saves time
- reduces human error
- aids early diagnosis
- optimises resources

feature extraction

Both PSD & wavelet transform

PSD - Helps understand how signal energy is distributed across frequencies

wavelet trans form - Capture short term pattern in EEG signals, which are crucial.

Advanced hyperparameter tuning (Grid Search)

Random forest - no. of estimators, tree depth, & sample splits.

SVM → kernel type, cvalue & gamma tuning

decision tree - max depth

why not deep learning $\rightarrow$ CNN & LSTM.

requires more dataset $\rightarrow$ Not available

why use both PSD & wavelet transform?

EEG signal — non-stationary

Fourier transform fail because $\rightarrow$ they assume a signal's freq components const.

PSD — helps analyze how signal energy is distributed across diff. freq bands

wavelet transform $\rightarrow$ captures both freq & time-localized changes

use $\rightarrow$ Welch's method $\rightarrow$ piche likha hai

Discrete wavelet transform

decomposes signal into diff. freq bands while preserving time info.

pywt wavelet

— break down the EEG signal into multiple levels

— approximate coeff $\rightarrow$ captures general shape of signal (approximate & detail coeff.)

— detail coeff $\rightarrow$ capture rapid change

model is trained using PSD features $\rightarrow$ compact & reduced dimens. ^{onset}

DWT $\rightarrow$ visual confirmation & potential feature enhancement

Jaise - Stand alone
aur

connectivity

Students

name,
Email,
pwd
Studentid.
role

user

Teacher

190 - soft
1470 - hard
745 - maced
1400 - rutile
180 - vadapav

Siddhi

10

11

75

20

45

+

18

At 11:9
→